

WOREX

WORLD RESOURCE & EXCHANGE LTD

FUNDING PROPOSAL — APRIL 2026

IoT-AI Smart Dairy Farm Management Platform

*Enhancing dairy productivity, reproductive efficiency, and
environmental sustainability across Eastern & Northern Rwanda
through the **MooMe** platform.*

6

DISTRICTS

500+

FARMERS

12

MONTHS (YEAR 1)

\$205K

YEAR 1 BUDGET

THE CHALLENGE

Rwanda's Dairy Sector Operates Far Below Potential



Low Milk Yields

Average 1 - 3.6 L/cow/day vs. genetic potential of 20–50 L for improved breeds.

85% BELOW POTENTIAL



Disease Prevalence

60% mastitis prevalence. East Coast Fever causes 1.1M cattle deaths/year in Africa.

60% MASTITIS RATE



Reproductive Failure

AI success rates below 40%. Over 95% of farmers keep no reproductive records.

<40% CONCEPTION



High Emissions

Emission factors 3–26x above IPCC defaults. Manure energy potential untapped.

3–26X ABOVE BASELINE



Feed Scarcity

Unbalanced feeding, no dry-season conservation. Land subdivision shrinks grazing areas.

CRITICAL GAP



Existing Solutions Too Expensive

International IoT platforms cost \$2000 per farm — unaffordable for smallholders.

\$2000 / FARM

OUR SOLUTION

Introducing MooMe — An Affordable AI-IoT Platform

MooMe is an AI-powered, IoT-integrated dairy farm management mobile application — purpose-built for smallholder farms. It runs on standard Android smartphones, integrates with low-cost IoT kits (**to reduce between 40% to 60%** vs. \$2,000/farm for alternatives), and delivers actionable intelligence through an intuitive mobile interface.

01

Smart Feed Management

IoT sensors + ML-optimized rations
with sweet potato silage

02

Disease Early Warning

Wearable sensors with <2hr alert
response capability

03

Reproductive Management

AI-powered estrus detection & digital
records

04

Biodigester Monitoring

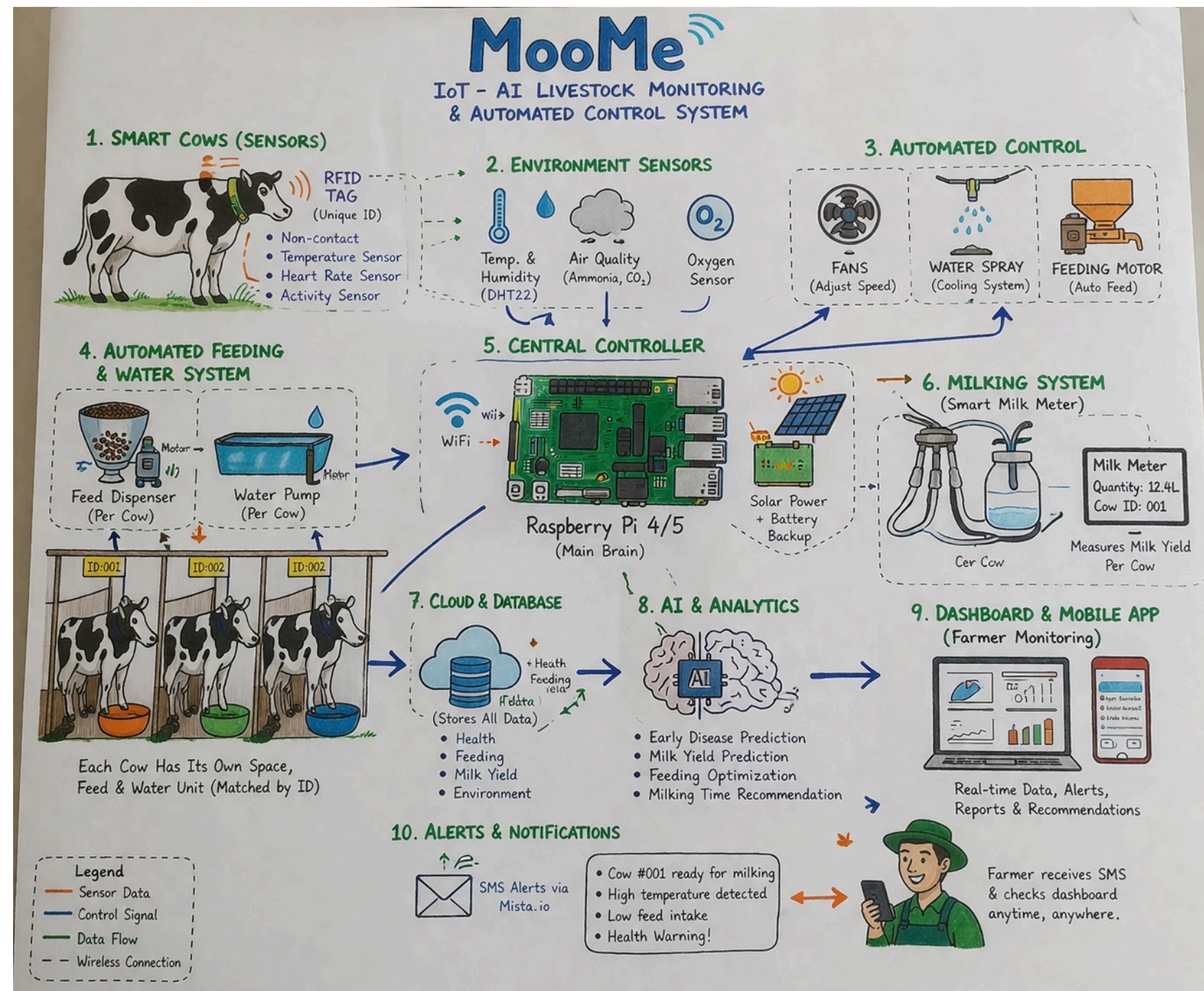
ML-predicted biogas output &
emissions tracking

05

Unified Platform

Cloud dashboard + SMS/USSD for all
literacy levels

MooMe – IoT-AI Livestock Monitoring & Automated Control

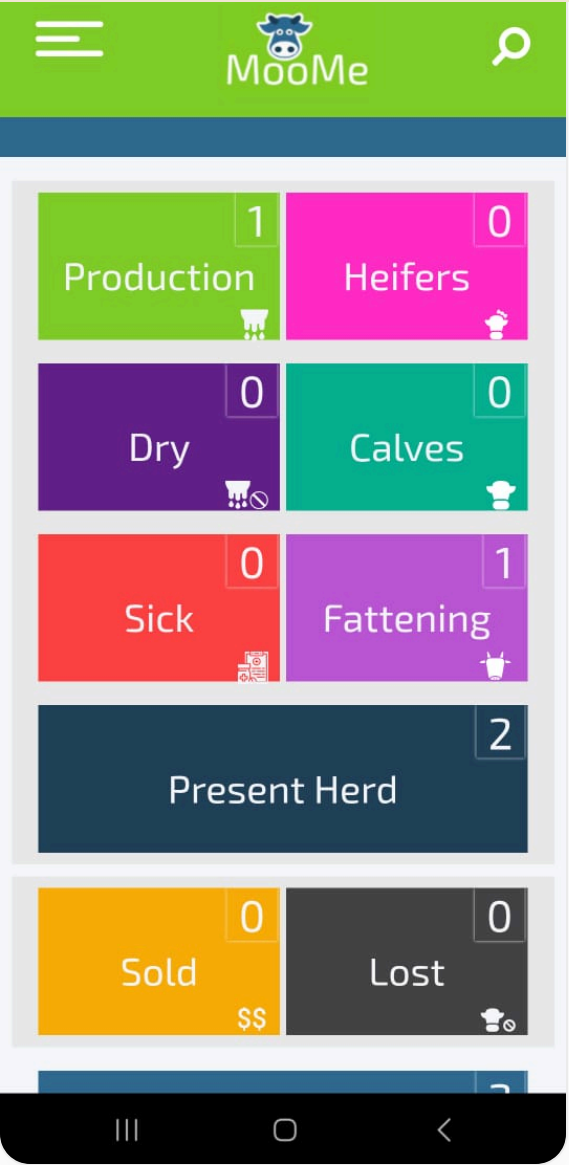


10 integrated subsystems: smart cow sensors (RFID, temperature, heart rate, activity), environment sensors, central Raspberry Pi controller, automated feeding & water, milking system, cloud database, AI analytics, dashboard & mobile app, and SMS alerts — all solar-powered.



Dashboard

Farm analytics overview



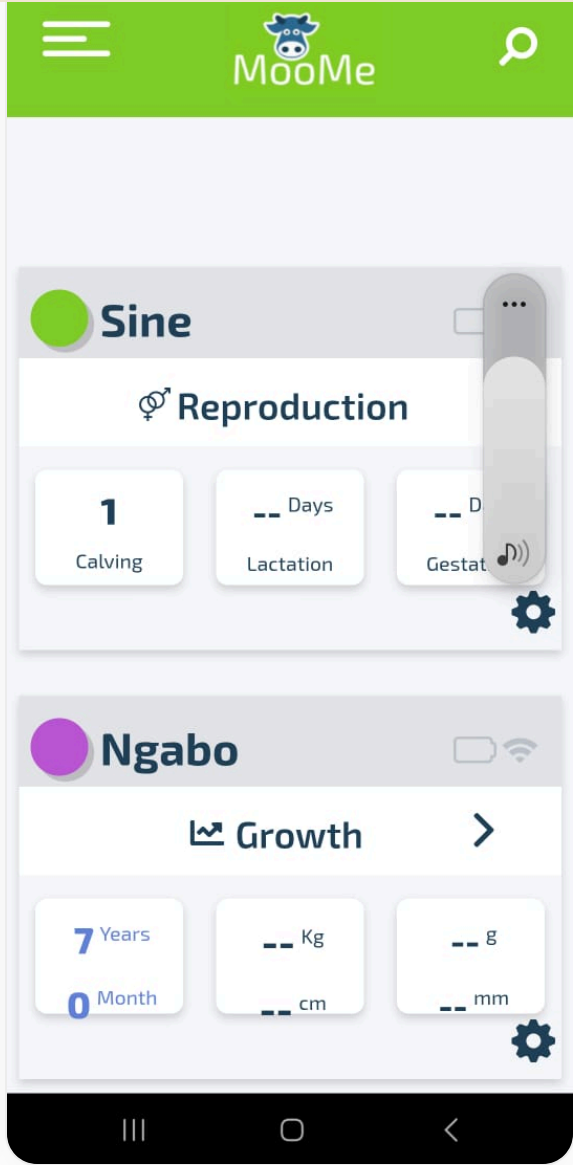
Herd Management

Real-time herd composition



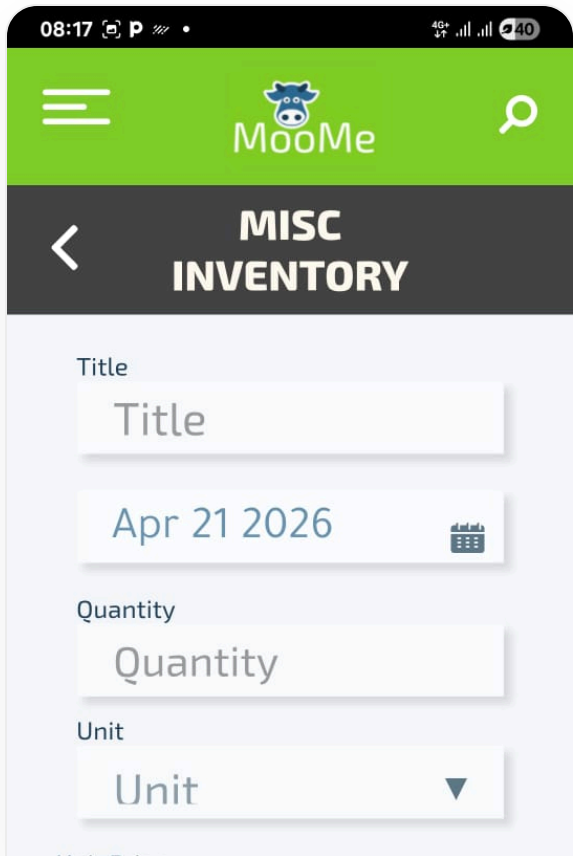
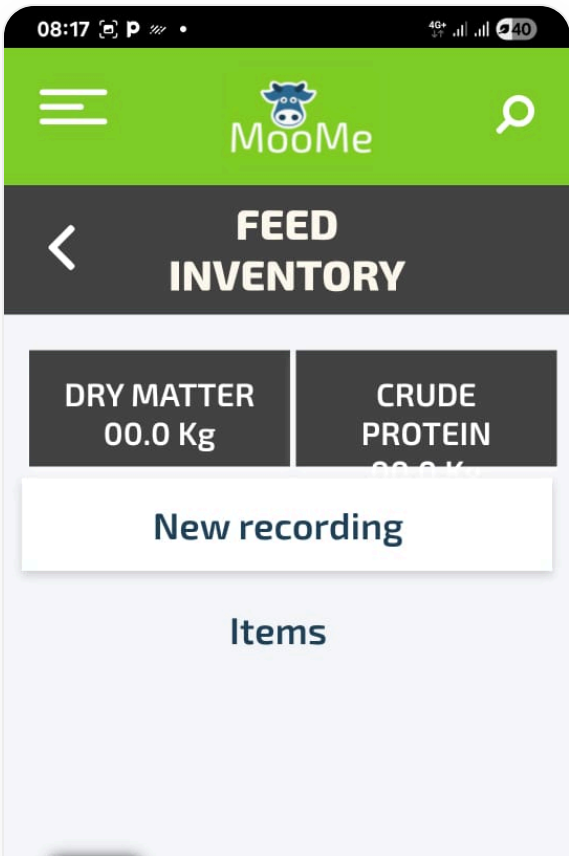
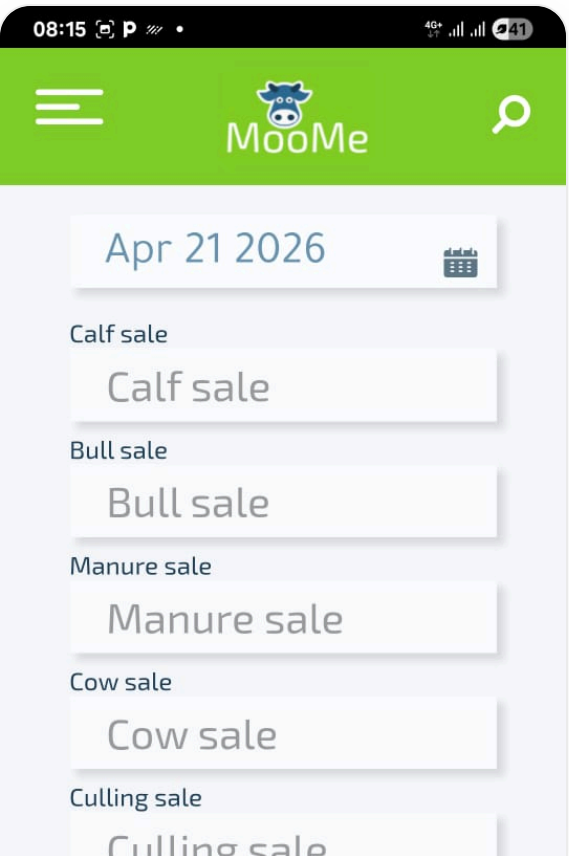
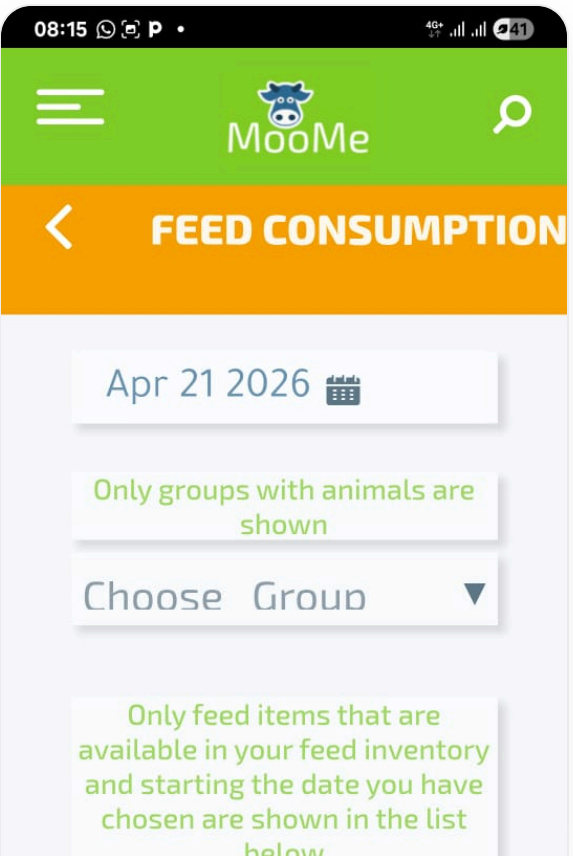
Milk Production

Daily recording & RWF income



Reproduction

Calving, lactation & gestation



PROVEN TRACK RECORD

Validated Across Africa — Not a Prototype

 Kenya Smallholder dairy farms in Rift Valley; mixed agro-pastoral systems IoT sensor accuracy for milk yield; mobile app adoption by low-literacy farmers	 Uganda Cooperative dairy clusters; Western Uganda highlands Group-based feed management; multi-farm data aggregation validated	 Nigeria Peri-urban commercial farms; Kaduna & Plateau states Economic module: income tracking, cost analysis, profitability reporting	 Tunisia Technology hub; MooMe core engineering & expert base Full platform development, AI model training, IoT hardware calibration
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COMPETITIVE ADVANTAGE

MooMe vs. Existing High-Cost Solutions

CRITERIA	EXISTING HIGH-COST SYSTEMS	MOOME (PROPOSED)
Hardware Cost	\$2,000 per farm	~\$1,000 per farm (4–6x cheaper)
Connectivity	Requires stable internet & cloud servers	Mobile-first, works on basic 4G
Language Support	Mostly English / European languages	Multilingual incl. Kinyarwanda
Target Market	Large commercial farms	Smallholder & mid-scale farms
Reference Countries	Europe, North America	Kenya, Uganda, Nigeria, Tunisia
Local Support	No in-country training capacity	3-day expert training + 6 local technicians

THE \$1,000 PER-FARM COST = 4× TO 6× COST REDUCTION VS. INTERNATIONAL ALTERNATIVES

EXPECTED IMPACT

Measurable Outcomes That Transform Livelihoods

+10% - 15%

Milk yield increase
across target farms

>10% - 15%

AI conception rate
up from <25% baseline

-15%

GHG emissions
per farm within 3 years

85%+

Disease detection
accuracy (<2hr alerts)

200+

Active farmers on
platform by Year 3

50+

Youth and women empowered as
IoT technicians & data agents

Immediate (Year 1)

- 12 farms operating MooMe IoT kits
- 6 trained local technicians in Rwanda
- Validated IoT data accuracy benchmarks
- App localized for Rwanda (Kinyarwanda)

Long-Term (Years 2–5)

- Scale to 500+ farms nationally
- Rwanda as regional MooMe hub for EAC
- Integration with national dairy programs
- Carbon credit revenue & financial sustainability

IMPLEMENTATION PLAN

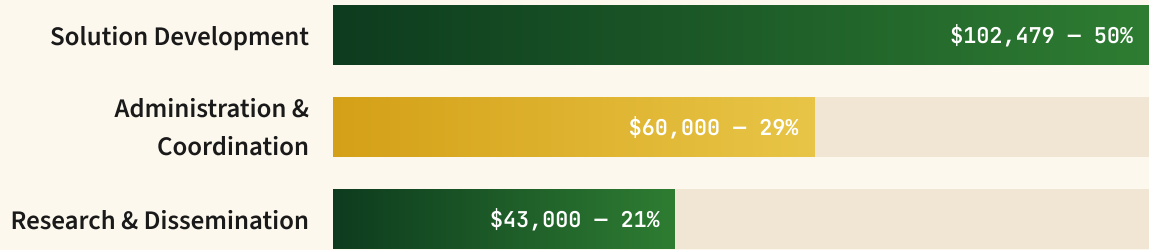
12-Month Year 1 Deployment Roadmap



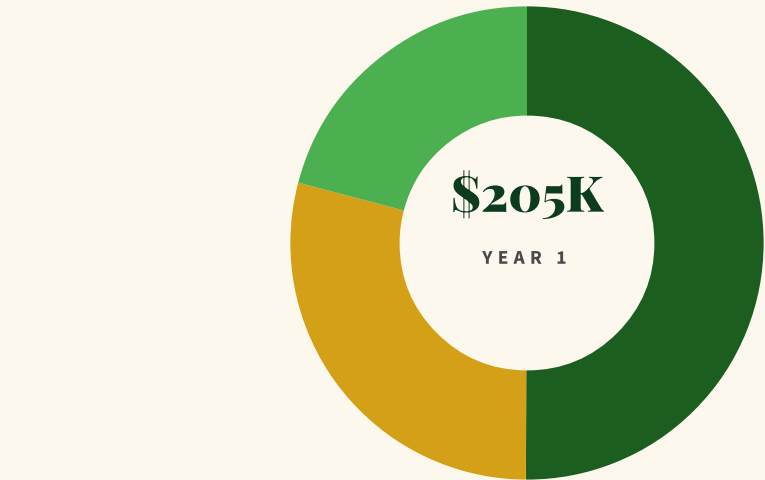
YEAR 1 INVESTMENT

\$205,479

Year 1 Budget (USD)



\$1,000 / farm IoT Hardware Kit (50 pilot farms)	\$24,000 Total Hardware Investment
Cloud + AI Platform & ML Infrastructure	Training 6 technicians + 50 farmers



Development (50%) Administration & Coordination (29%) Research & Dissemination (21%)

STRATEGIC PARTNERSHIPS

A Multi-Stakeholder Consortium for Impact

WOREX Ltd. Lead Implementer — Project coordination, design & delivery LEAD	ACTION FAMILY TECH LTD Technology Partner — Web app, IoT & AI systems TECHNOLOGY	MooMe (Tunisia) Platform Provider — Expert training & knowledge transfer PLATFORM
FAO Rwanda Funding — Climate-smart agriculture alignment UN AGENCY	Rwanda Agriculture Board (RAB) Co-implementing — Extension services & data sharing GOVERNMENT	MINAGRI Policy Oversight — NST2, NDCs, agricultural policy GOVERNMENT
University of Rwanda & University of Kigali Research — College of Agriculture & Veterinary Medicine RESEARCH		

Aligned with **NST2 (2024–2029)**, Rwanda's **NDCs** under the Paris Agreement, **FAO's climate-smart agriculture** agenda, and the **AU Agenda 2063**.



FUNDING REQUEST

USD \$205,479

Year 1 Investment — Design, Development & Pilot Deployment

EMAIL

worexltd2010@gmail.com

PHONE

+250 788 300 975

ADDRESS

Nyagatare Sector,
Nyagatare City

*"Resources are meant to be shared. When shared, they become
catalysts for development."*

WOREX LTD. • WORLD RESEARCH & EXTENSION • SUBMITTED APRIL 2026 • 12-MONTH YEAR 1 FUNDING

60%+ WOMEN BENEFICIARIES • 50+ YOUTH JOBS • SOLAR-POWERED • CARBON CREDIT ELIGIBLE • 4-6X CHEAPER THAN
ALTERNATIVES